Project
PORTFOLIO

HI, I AM SHOFA

Data & Machine Learning Enthusiast

As a student of Information Systems at Surabaya State University, my primary focus lies in honing my expertise within the realms of Data Analytics and Machine Learning.
I am keen on broadening my knowledge base and initiating a career journey within the expansive realm of data.

TOOLS

Python

Tableau

Visual Studio Code

Google Colab

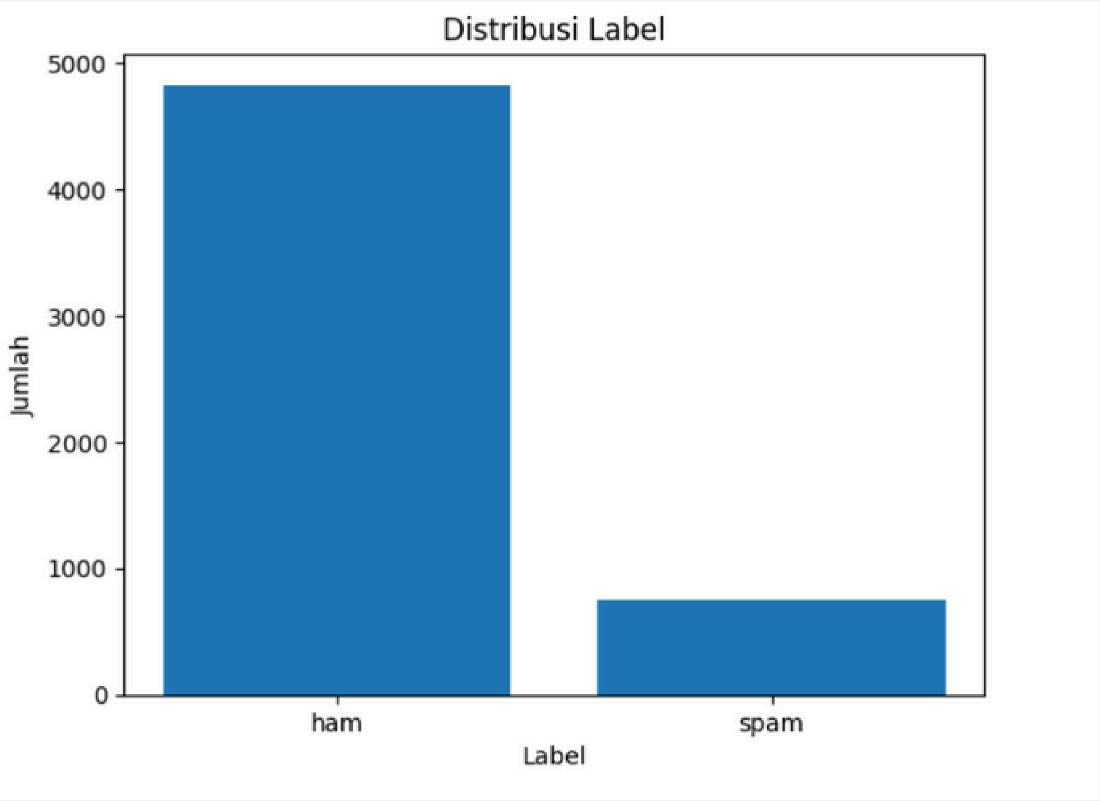
SPSS

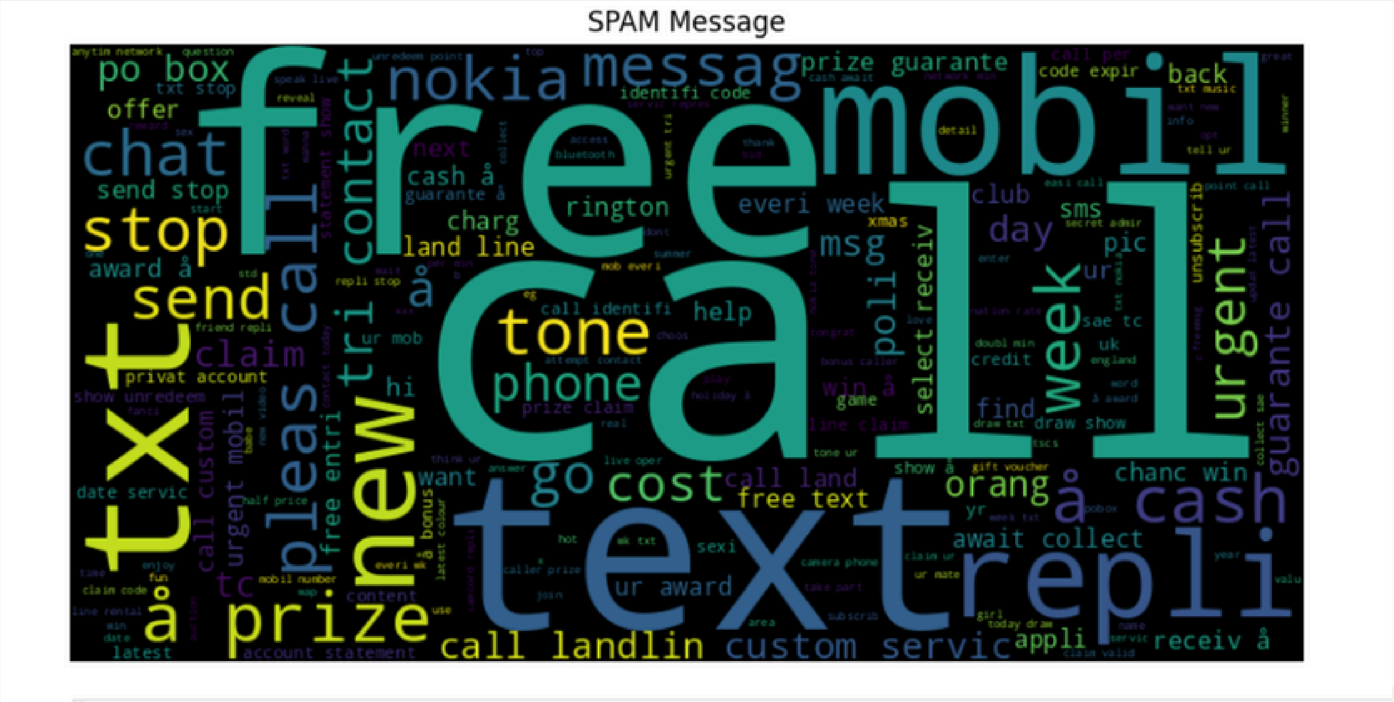
Ms. Excel

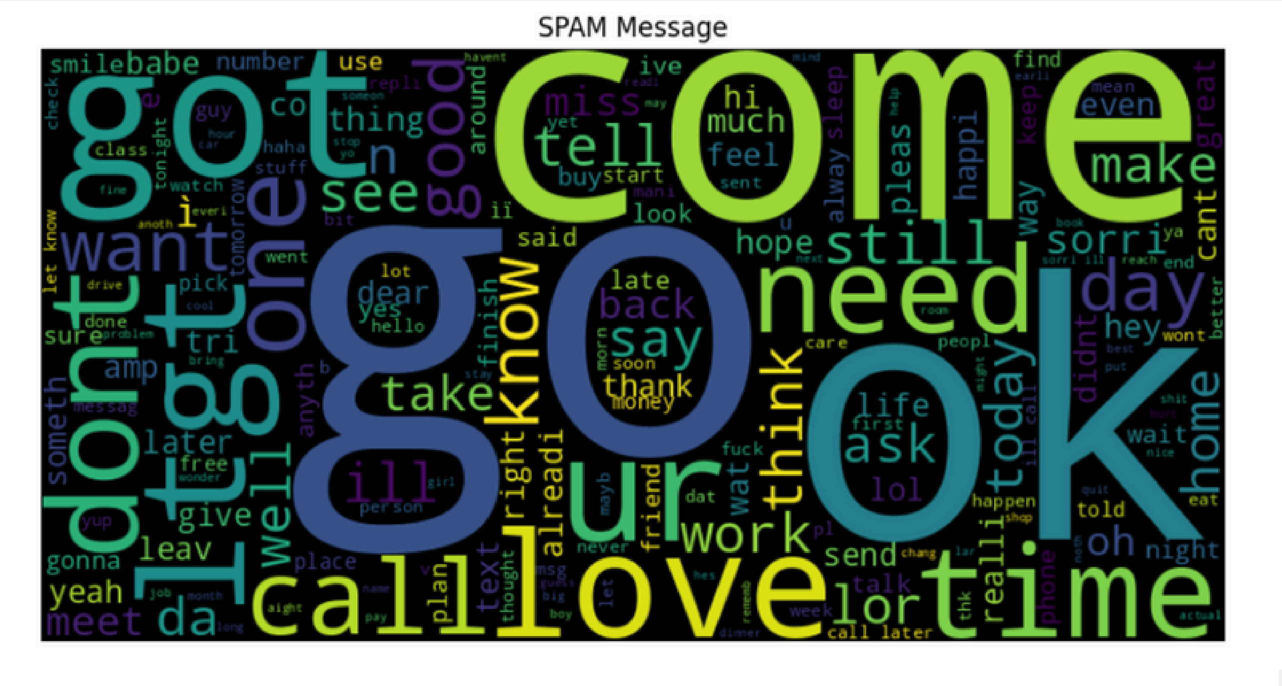
RELEVANT PROJECTS

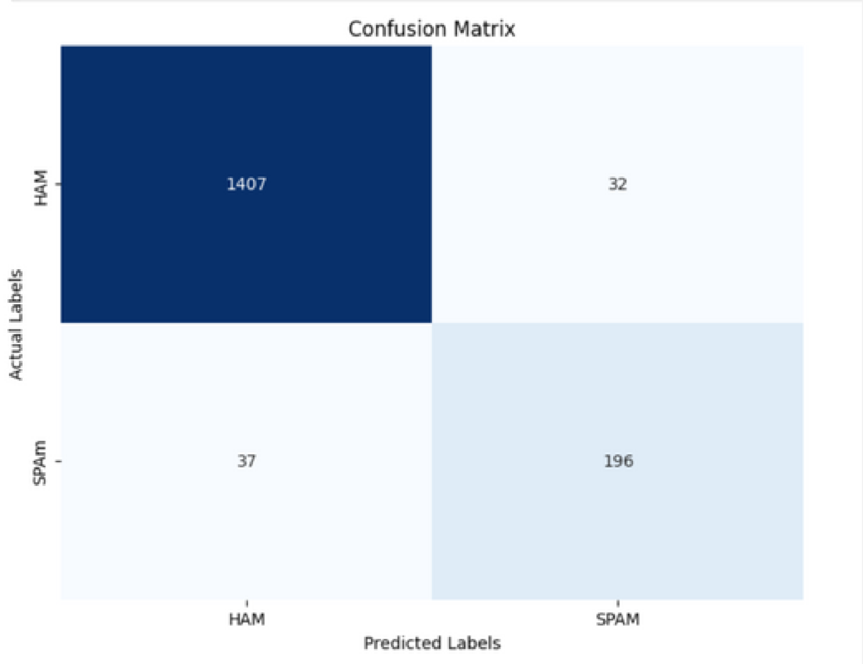
SMS SPAM CLASSIFICATION

This model is created by implementing the LSTM (Long Short-Term Memory) algorithm with high accuracy to predict whether an SMS is spam or not. The built model's accuracy is very high, reaching up to 96%.









Link

STROKE PREDICTION

This predictive model was developed using the Random Forest algorithm and leveraged hyperparameter tuning to achieve the best possible model accuracy, with the highest accuracy reaching 96%. Data analysis revealed that age, hypertension, and blood sugar levels are the primary causes of stroke.

Random Search

In [37]:
```
from sklearn.model_selection import RandomizedSearchCV

n_estimators = [int(x) for x in np.linspace(start=200, stop=2000, num = 5)]
max_depth = [int(x) for x in np.linspace(10, 1000, 5)]
min_samples_split = [2, 5, 10, 14]
min_samples_leaf = [1, 2, 4, 6, 8]
random_search_params = {'n_estimators': n_estimators,
                        'max_depth':max_depth,
                        'min_samples_split':min_samples_split,
                        'min_samples_leaf':min_samples_leaf}
random_search_params
```

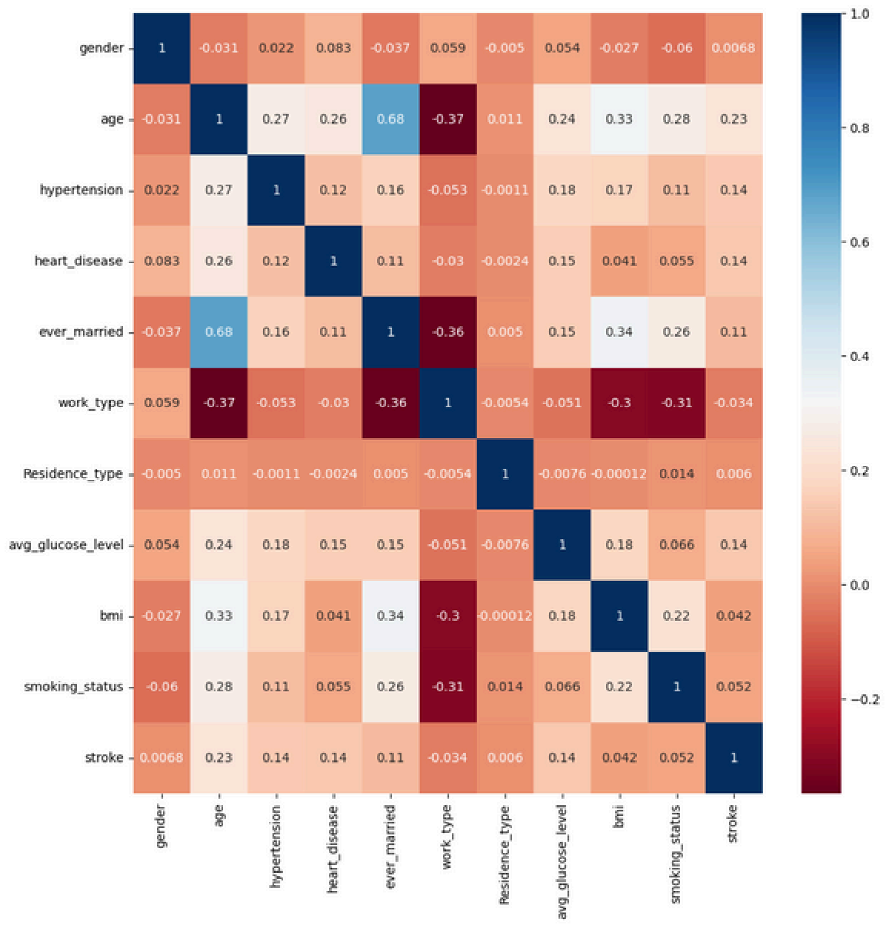
Out[37]:
```
{'n_estimators': [200, 650, 1100, 1550, 2000],
 'max_depth': [10, 257, 505, 752, 1000],
 'min_samples_split': [2, 5, 10, 14],
 'min_samples_leaf': [1, 2, 4, 6, 8]}
```

In [ ]:

In [38]:
```
rf_randomcv = RandomizedSearchCV(estimator=RandomForestClassifier(),
                                 param_distributions=random_search_params,
                                 n_iter=20,
                                 cv=5,
                                 random_state=46,
                                 n_jobs=-1, #parallel processing
                                 scoring='f1',
                                 verbose=3)
rf_randomcv.fit(x_train, y_train)
```

```
Fitting 5 folds for each of 20 candidates, totalling 100 fits
```

Out[38]:
```
RandomizedSearchCV(cv=5, estimator=RandomForestClassifier(), n_iter=20,
                   n_jobs=-1,
                   param_distributions={'max_depth': [10, 257, 505, 752, 1000],
                                        'min_samples_leaf': [1, 2, 4, 6, 8],
                                        'min_samples_split': [2, 5, 10, 14],
                                        'n_estimators': [200, 650, 1100, 1550,
                                                         2000]},
                   random_state=46, scoring='f1', verbose=3)
```



In [42]:
```
prediction_rf_rcv = rf_randomcv_best.predict(x_train)
accuracy_rf_rcv = accuracy_score(y_train, prediction_rf_rcv)
print(f"Random Forest Train Accuracy: {accuracy_rf_rcv}")
predictions_rf_rcv = rf_randomcv_best.predict(x_test)
accuracy_rf_rcv = accuracy_score(y_test, predictions_rf_rcv)
print(f"Random Forest Test Accuracy: {accuracy_rf_rcv}")
```

```
Random Forest Train Accuracy: 0.9685681024447031
Random Forest Test Accuracy: 0.9511201629327902
```

Link

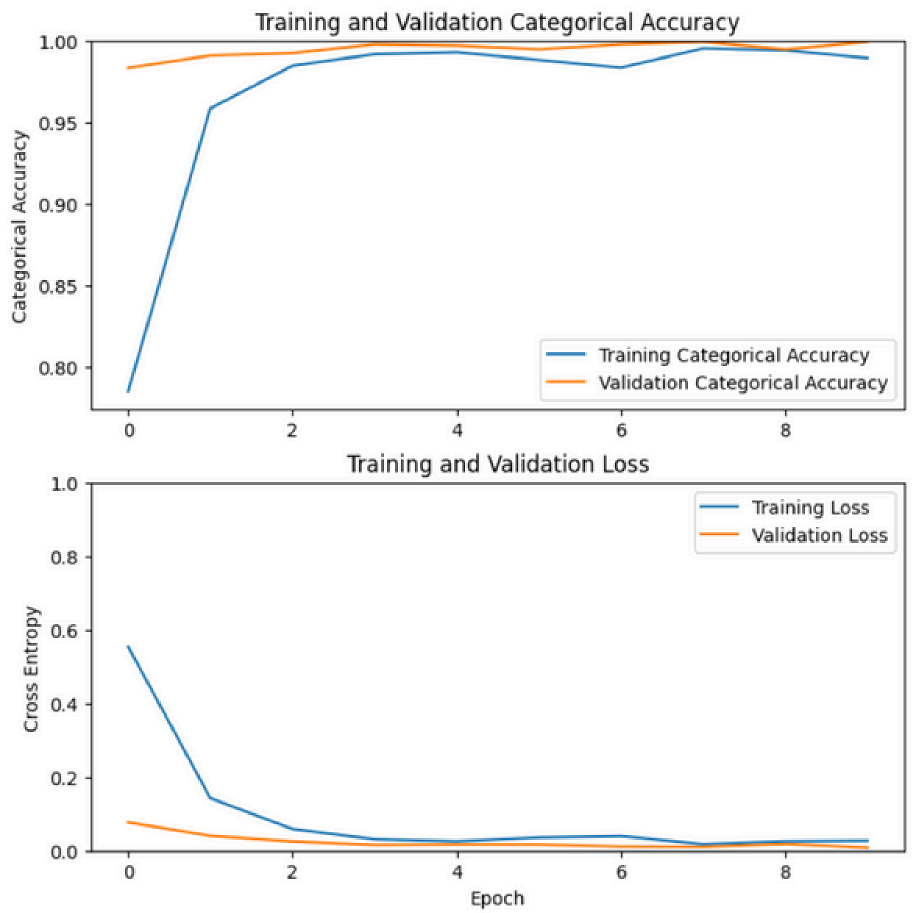
IMAGE DETECTIONS

This model enables users to automatically recognize and classify images of rock, paper, and scissors with high accuracy. It can be utilized to implement technology in the gaming world.

```
Saving Rock-paper-scissors_(scissors).png to Rock-paper-scissors_(scissors).png
```

```
1/1 [==============================] - 0s 55ms/step

 Ini adalah scissors
```

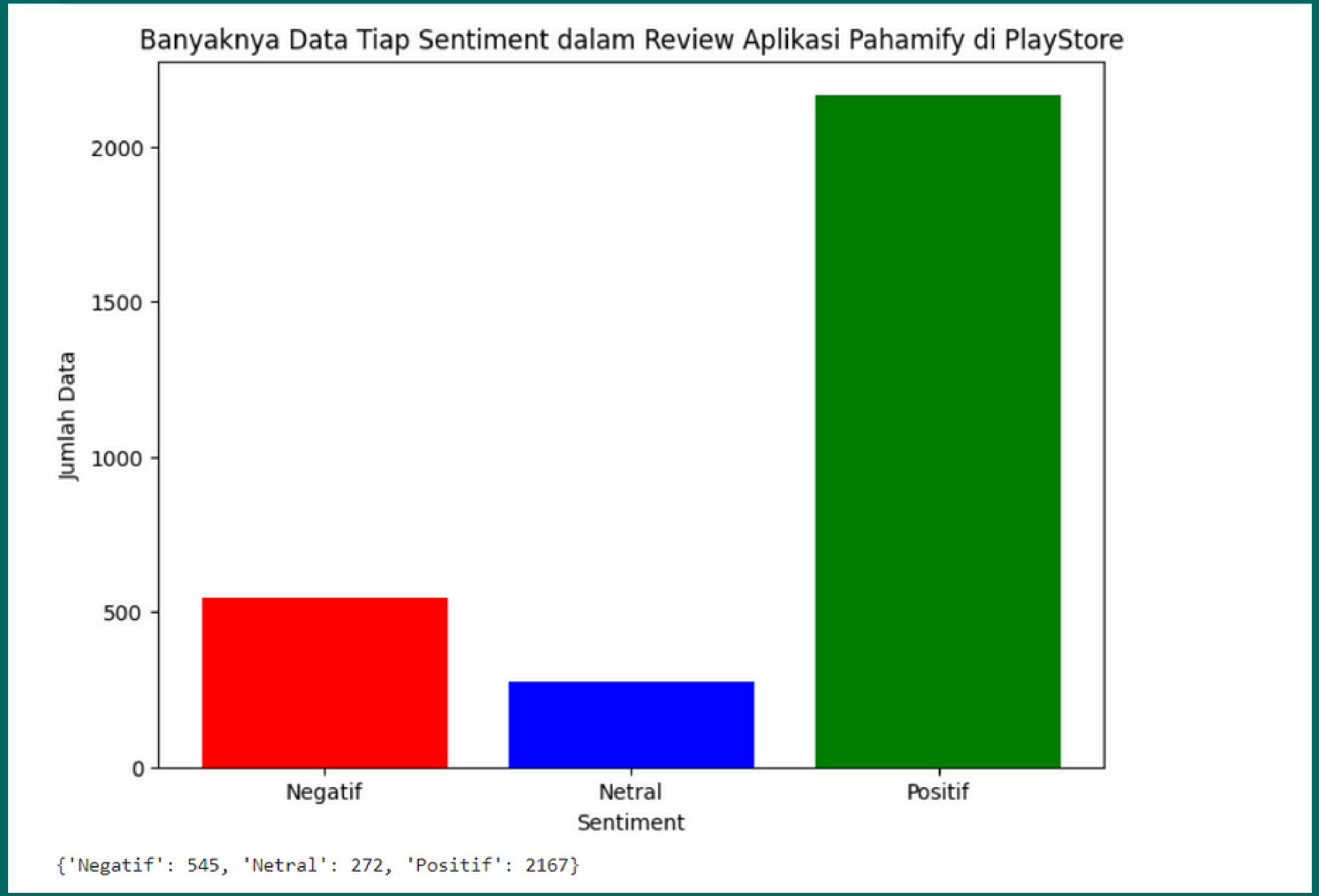
```
In [ ]: model.compile(
            optimizer=tf.keras.optimizers.Adam(learning_rate=1e-3),
            loss='categorical_crossentropy',
            metrics=['accuracy']
        )

In [ ]: history = model.fit(train_generator, validation_data=validation_generator,
                            steps_per_epoch=train_generator.n // train_generator.batch_size,
                            validation_steps=validation_generator.n // validation_generator.batch_size,
                            epochs=10)

Epoch 1/10
27/27 [==============================] - 135s 5s/step - loss: 0.5538 - accuracy: 0.7853 - val_loss: 0.0765 - val_accuracy: 0.
9832
Epoch 2/10
27/27 [==============================] - 131s 5s/step - loss: 0.1422 - accuracy: 0.9585 - val_loss: 0.0399 - val_accuracy: 0.
9909
Epoch 3/10
27/27 [==============================] - 132s 5s/step - loss: 0.0586 - accuracy: 0.9846 - val_loss: 0.0247 - val_accuracy: 0.
9924
Epoch 4/10
27/27 [==============================] - 131s 5s/step - loss: 0.0306 - accuracy: 0.9917 - val_loss: 0.0151 - val_accuracy: 0.
9977
Epoch 5/10
27/27 [==============================] - 130s 5s/step - loss: 0.0251 - accuracy: 0.9929 - val_loss: 0.0163 - val_accuracy: 0.
9970
Epoch 6/10
27/27 [==============================] - 106s 4s/step - loss: 0.0352 - accuracy: 0.9881 - val_loss: 0.0160 - val_accuracy: 0.
9947
Epoch 7/10
27/27 [==============================] - 131s 5s/step - loss: 0.0393 - accuracy: 0.9834 - val_loss: 0.0108 - val_accuracy: 0.
9977
Epoch 8/10
27/27 [==============================] - 128s 5s/step - loss: 0.0173 - accuracy: 0.9953 - val_loss: 0.0098 - val_accuracy: 0.
9992
Epoch 9/10
27/27 [==============================] - 131s 5s/step - loss: 0.0249 - accuracy: 0.9941 - val_loss: 0.0175 - val_accuracy: 0.
9947
Epoch 10/10
27/27 [==============================] - 104s 4s/step - loss: 0.0268 - accuracy: 0.9893 - val_loss: 0.0078 - val_accuracy: 0.
9992
```



Link

SENTIMENT ANALYSIS

The Pahamify app is an online learning platform designed to enhance students' understanding of their study materials. The data used was obtained from Google Play through data scraping, limited to 3,000 data points. The primary goal of NLP (Natural Language Processing) is to enable computers to understand, interpret, and respond to human language in a natural way. Based on the scraping results, the Pahamify app shows excellent reviews, with a total of 72% of users expressing satisfaction with the app.



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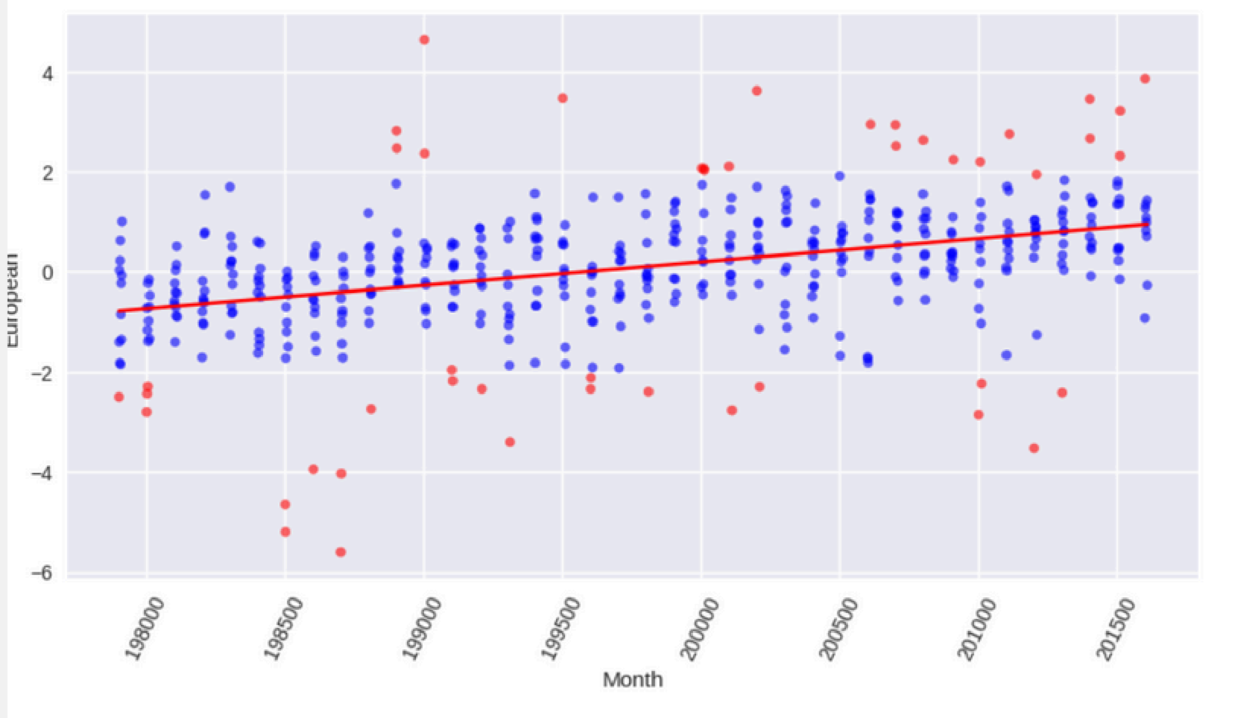
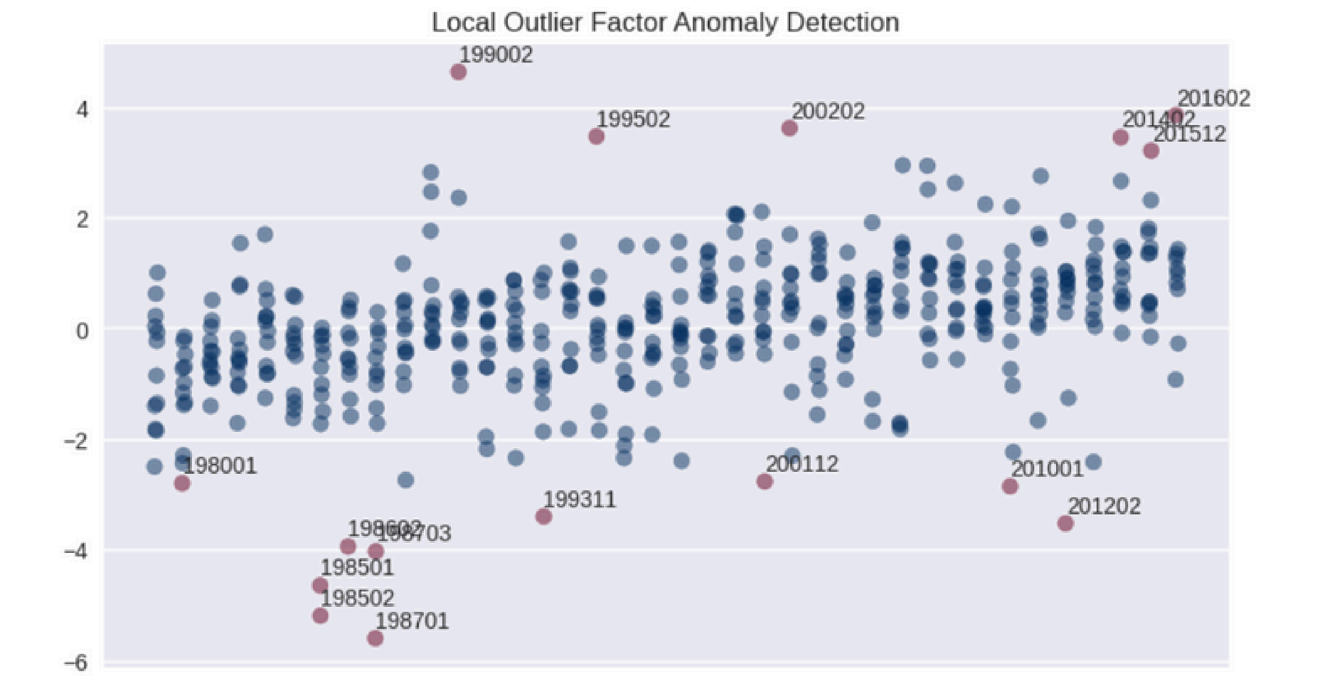
ANOMALY DETECTIONS

The goal of building this model is to detect anomalies or outliers within the dataset. The model was developed using several algorithms, and the highest accuracy was achieved by the Local Outlier Factor (LOF) algorithm with the default n-neighbors (20):

- Accuracy: 91.41%
- Precision: 92.89%
- Recall: 91.41%
- F1 Score: 92.15%





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CUSTOMER SEGEMNTATION

The model generated 5 clusters, as follows:

- Cluster 0: Average age is 43 years, annual income is 93.28, and mall spending score is 20.64.
- Cluster 1: Average age is 25 years, annual income is 25.69, and mall spending score is 80.53.
- Cluster 2: Average age is 33 years, annual income is 87.11, and mall spending score is 82.66.
- Cluster 3: Average age is 25 years, annual income is 40.25, and mall spending score is 60.91.
- Cluster 4: Average age is 32 years, annual income is 86.04, and mall spending score is 81.66.

Cluster	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	2.0	25.857143	26.642857	79.142857
1	1.0	39.500000	85.150000	14.050000
2	2.0	54.080000	53.240000	49.520000
3	1.0	32.947368	86.052632	81.263158
4	1.0	58.115385	48.038462	41.269231

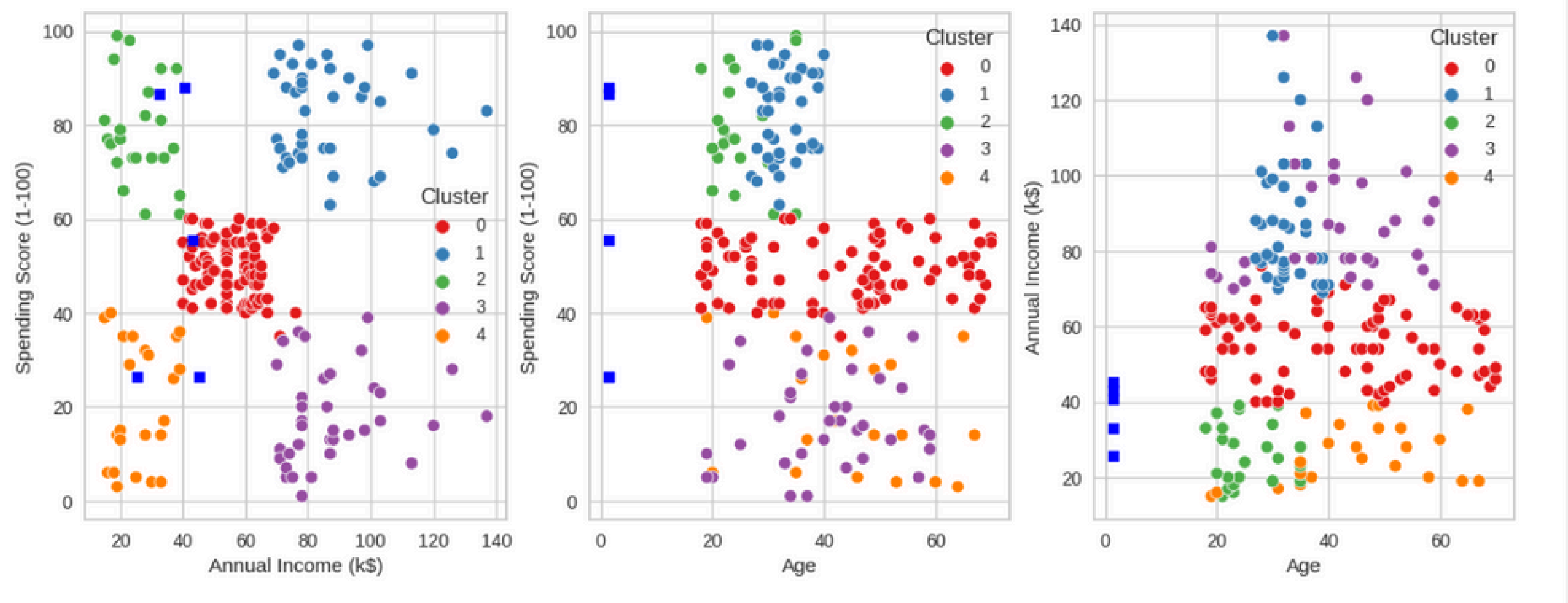

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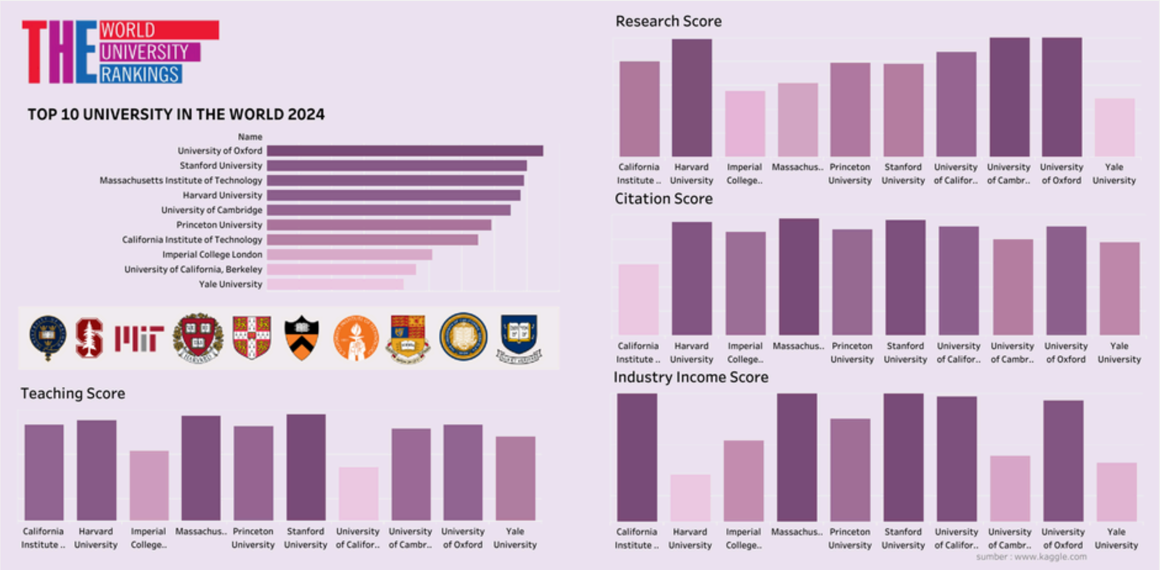
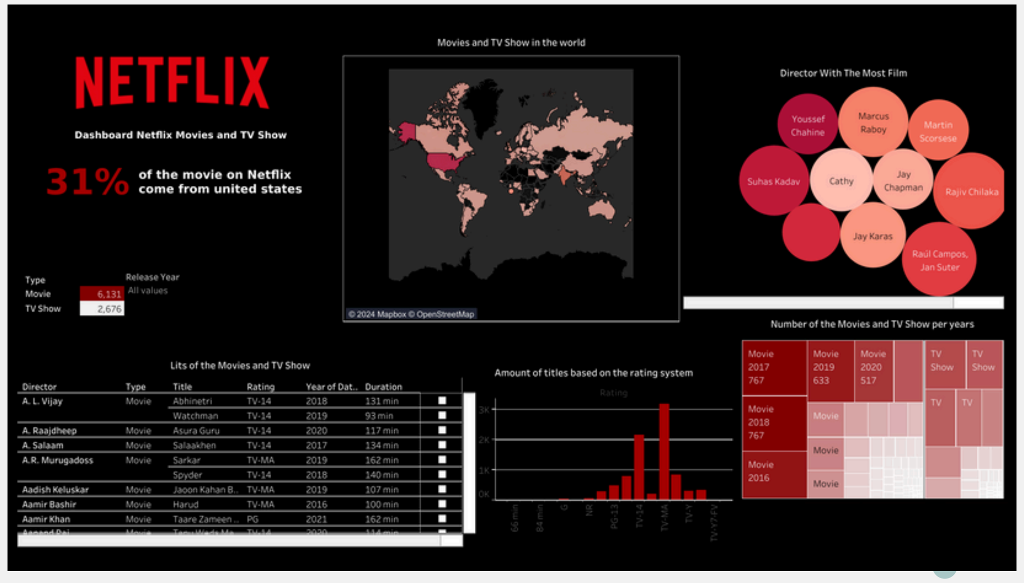
TABLEAU VISUALISATION

Amazon Sales Dashboard



THE WORLD University Ranking



Netflix Dashboard

[Link](#)

THANK YOU!